React Prototype with Components
Whisper Notes

Introduction

This document provides an overview of our current design and architectural decisions, including both the parts that are already implemented and the parts that still need to be completed. The explanations are structured according to the page sections of the frontend. In addition, this document describes the features that remain consistent with the previously submitted design draft, as well as the design changes that were introduced later.

Conceptual Changes After the Design Draft

- After the design draft, we decided to make several changes to our original design concept. Instead of using the blue color tones shown in the mockups, we chose a more minimalist and simple color palette based on green tones. This decision was made to create a calmer visual appearance and to make the interface less tiring for the user's eyes.
- We also changed the navigation flow after login. In the design draft, the user was planned to be redirected directly from the login page to the transcribe page. Instead, we decided to add a separate home page. After logging in, the user is redirected to the home page. The home page provides an overview of the user's activity, including recent transcripts, total processed audio time, shared files, and quick navigation to the main application pages.

Login-Home Page

The Login and Home pages define the main entry flow of the application. The project uses `react-router-dom` routes with protected pages. If the user is not logged in, protected routes redirect to the Login page. After a successful login, the user is redirected to the Home page.

- **Login form:** The Login page contains email and password fields. Both fields must be filled in before the user can continue. If one of them is empty, an error message is shown. The current version uses demo login data and does not yet validate the credentials against a backend.
- **Authentication state:** After login, the application stores the current user information in `localStorage`. This allows the user session to remain available after a page refresh. The entered email is normalized, and the displayed user name is generated from the email address.
- **Navigation after login:** Before login, the navigation only shows the Login option. After login, the navigation changes and shows Home, Transcribe, Results, Shared Files, and a Sign out button. Signing out clears the stored user state and returns the application to the unauthenticated flow.
- **Home dashboard:** The Home page works as a dashboard after login. It shows the user profile information, including name, email, role, and group. It also provides direct navigation cards for creating a new transcription, reviewing results, and opening shared files.
- **Activity overview:** The Home page summarizes available transcript data. It displays the number of saved transcription results, the total processed audio time, the latest result, the number of shared files, the total shared audio time, and the latest shared file. These values are currently calculated from local frontend demo data.
- **Scope:** The login and home flow is implemented as a frontend prototype. A real backend authentication system, user accounts, password verification, and server-side session handling are planned for a later development step.

Transcribe Page

The Transcribe page represents the planned area where users will create a new note from audio. In the current React prototype, this page is implemented as a placeholder view with a title and a short description explaining that users will be able to upload or record audio and turn spoken words into organized notes.

- **Current implementation:** The page is already connected to the application navigation and protected route structure, so only logged-in users can access it.

- **Planned upload and recording flow:** In a later version, this page should allow the user to upload an audio file or record audio directly in the browser.
- **Planned transcription settings:** Future versions should include options such as language selection, timestamp generation, speaker detection, and engine selection.
- **Scope:** At this stage, the Transcribe page is a frontend placeholder. The actual transcription request, audio processing, and backend connection still need to be implemented.

Results Page

The Results page is used to review saved transcription results. It contains a sidebar with available transcript files and a main workspace for the selected result. The selected transcript can also be opened through a query parameter, so different saved results can be displayed with separate URLs.

- **Saved results list:** The sidebar lists the available transcription results. Each item shows the transcript title, group, language, and audio duration. Selecting an item updates the main workspace without leaving the Results page.
- **Transcript workspace:** The main area shows the selected file title, creator, group, and language. It displays the detected speakers, the audio progress area, the currently active transcript segment, and the full transcript list.
- **Speaker editing:** The page allows the user to edit the displayed speaker names directly in the interface. These names are stored in local React state and are used when showing, copying, or downloading the transcript text.
- **Audio playback and progress:** If an audio file is connected to the selected result, the user can play or pause the audio. The page shows the current timestamp, total duration, a visual waveform-style progress indicator, and a progress slider. The slider can be controlled by mouse or keyboard. If no audio file is connected, the page shows a message explaining that no audio file is available for that result yet.
- **Timestamp synchronization:** While the audio plays, the current time is tracked. The transcript segment that matches the current timestamp is highlighted and also shown in a separate current transcript area. This helps the user follow the text together with the audio.
- **Copy and download actions:** The user can copy the full transcript to the clipboard or download it as a `.txt` file. The exported transcript includes timestamps, speaker names, and transcript text.
- **Data source and scope:** The page currently uses local demo data, including a JSON transcript file and example saved results. Some text cleanup is handled in the frontend to fix character encoding issues in the sample transcript. Persistent saving, backend result storage, and real transcription history are planned backend integration tasks.

Shared Files Page

The Shared Files page allows users to review previously shared transcriptions. It follows the application's green color palette and rounded card layout.

- **Working features:** Users can select a shared file, play its audio through an interactive progress bar, and view the synchronized transcript segment.
- **Current data:** Transcriptions and audio files are loaded from local sample JSON data.
- **Design draft:** The original list-and-detail flow is preserved, while the visual style was adapted from blue to the current green design.
- **Incomplete work:** Backend integration, real user data, permissions, and search/filter features are not yet implemented.

Planned Backend / API

The current prototype uses a local JSON file and sample audio instead of the data structure originally planned in the design draft. In the next stage, the frontend should be connected to a backend using

WhisperLive. The backend will process audio, return transcription segments, and provide the data required by the Shared Files page.

AI tools were used to support the editing of this document, grammar improvement, and the preparation of the LaTeX code.